

TITLE: Log Analysis with Splunk

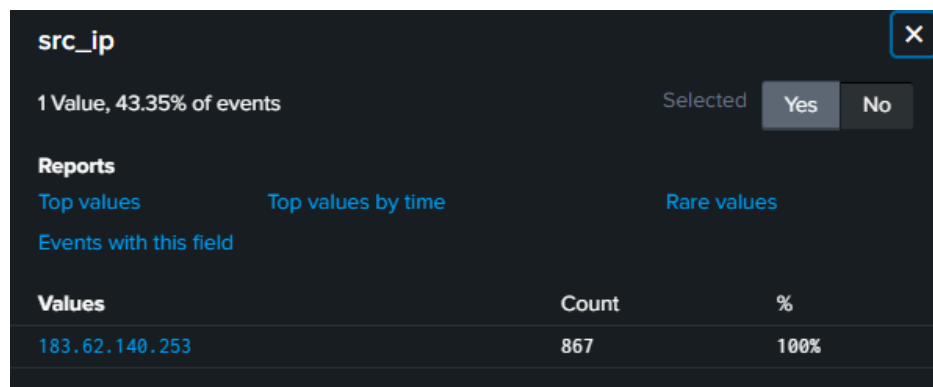
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SUMMARY

In this task, a log file of **2000** events were analyzed with Splunk. The analysis focused on using SPL to view authentication attempts events to derive anomalies and suspicious activities. An alert was configured after analysis for quick actions in future events.

ANALYSIS

To start with, I extracted some necessary fields including source IP, other IP addresses and time.



src_ip [X]

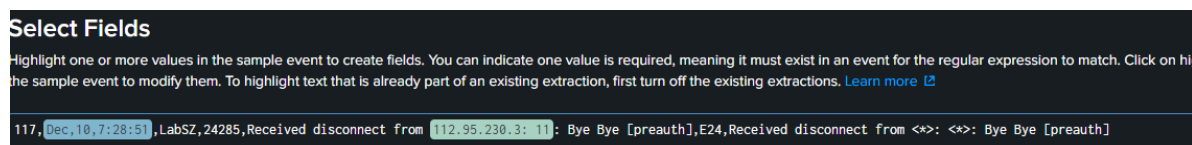
1 Value, 43.35% of events Selected Yes No

Reports

[Top values](#) [Top values by time](#) [Rare values](#)

[Events with this field](#)

Values	Count	%
183.62.140.253	867	100%

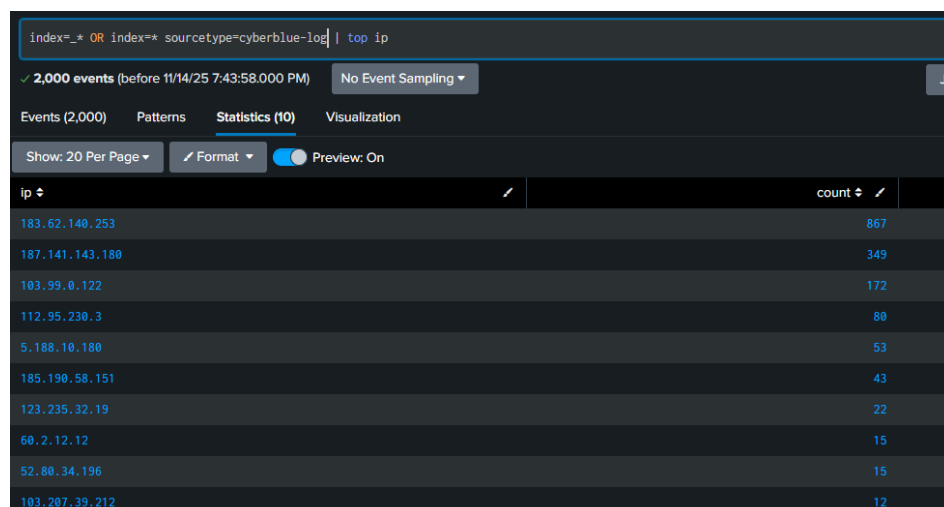


Select Fields

Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlights in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. [Learn more](#)

117,Dec,10,7:28:51,LabSZ,24285,Received disconnect from 112.95.230.3: 11: Bye Bye [preauth],E24,Received disconnect from <*>: <*>: Bye Bye [preauth]

Viewing the top IP addresses shows us the number of events that each unique address generated.



index=* OR index=* sourcetype=cyberblue-log | top ip

2,000 events (before 11/14/25 7:43:58.000 PM) No Event Sampling

Events (2,000) Patterns **Statistics (10)** Visualization

Show: 20 Per Page Format Preview: On

ip	count
183.62.140.253	867
187.141.143.180	349
103.99.0.122	172
112.95.230.3	80
5.188.10.180	53
185.190.58.151	43
123.235.32.19	22
60.2.12.12	15
52.80.34.196	15
103.207.39.212	12

After adding conditions to get **failed** or **invalid** events, we clearly see the number of times each IP tried to authenticate and FAILED. The IPs with high counts are extremely suspicious.

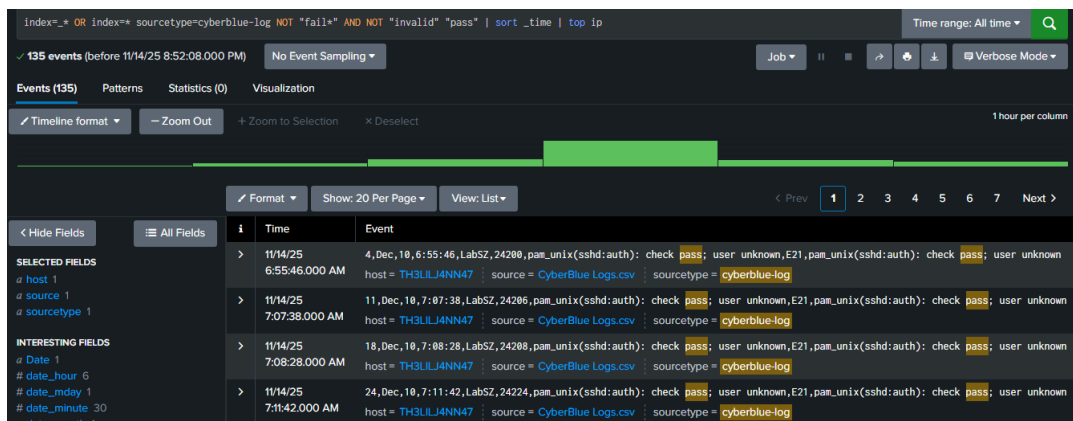
New Search	
index=* OR index== sourcetype=cyberblue-log "failed" OR "invalid" top ip	
836 events (before 11/14/25 7:56:00.000 PM) No Event Sampling Enable event sampling to run the search and return a random set of events. Job	
Events (836) Patterns Statistics (10) Visualization	
Show: 20 Per Page Format Preview: On	
ip	count
183.62.140.253	295
187.141.143.180	189
103.99.0.122	81
5.188.10.180	29
112.95.230.3	28
185.190.58.151	25
52.80.34.196	10
123.235.32.19	7
119.4.203.64	7
173.234.31.186	6

183.62.140.253 and 187.141.143.180 have the highest events. The images below tell us the authentication attempts are most likely automated.

i	Time	Event
>	11/14/25 10:54:27.000 AM	1020,Dec,10,10:54:27,LabSZ,24868,Invalid user zhangyan from 183.62.140.253,E13,Invalid user <*> from <*> host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log
>	11/14/25 10:54:29.000 AM	1026,Dec,10,10:54:29,LabSZ,24870,Invalid user dff from 183.62.140.253,E13,Invalid user <*> from <*> host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log
>	11/14/25 10:54:29.000 AM	1024,Dec,10,10:54:29,LabSZ,24868,Failed password for invalid user zhangyan from 183.62.140.253 port 33521 ssh2,E10,Failed password for invalid user <*> from <*> port <*> ssh2 host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log
>	11/14/25 10:54:31.000 AM	1030,Dec,10,10:54:31,LabSZ,24870,Failed password for invalid user dff from 183.62.140.253 port 33902 ssh2,E10,Failed password for invalid user <*> from <*> port <*> ssh2 host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log
>	11/14/25 10:54:33.000 AM	1033,Dec,10,10:54:33,LabSZ,24872,Failed password for root from 183.62.140.253 port 34263 ssh2,E9,Failed password for <*> from <*> port <*> ssh2 host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log

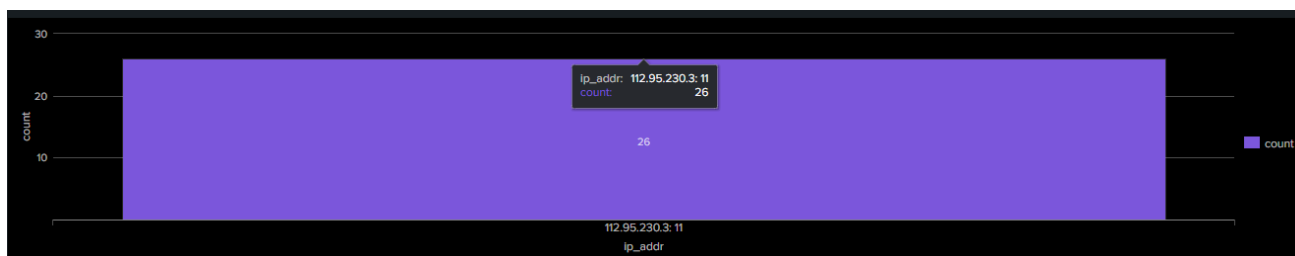
i	Time	Event
>	11/14/25 9:12:46.000 AM	517,Dec,10,9:12:46,LabSZ,24503,reverse mapping checking getaddrinfo for customer-187-141-143-180-sta.uninet-ide.com.mx [187.141.143.180] failed - POSSIBLE BREAK-IN ATTEMPT!,E27,reverse mapping checking getaddrinfo for <*> [<*>] failed - POSSIBLE BREAK-IN ATTEMPT! host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log
>	11/14/25 9:12:48.000 AM	519,Dec,10,9:12:48,LabSZ,24503,Failed password for root from 187.141.143.180 port 33314 ssh2,E9,Failed password for <*> from <*> port <*> ssh2 host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log
>	11/14/25 9:12:51.000 AM	521,Dec,10,9:12:51,LabSZ,24505,reverse mapping checking getaddrinfo for customer-187-141-143-180-sta.uninet-ide.com.mx [187.141.143.180] failed - POSSIBLE BREAK-IN ATTEMPT!,E27,reverse mapping checking getaddrinfo for <*> [<*>] failed - POSSIBLE BREAK-IN ATTEMPT! host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log
>	11/14/25 9:12:53.000 AM	523,Dec,10,9:12:53,LabSZ,24505,Failed password for root from 187.141.143.180 port 34508 ssh2,E9,Failed password for <*> from <*> port <*> ssh2 host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log
>	11/14/25 9:12:57.000 AM	529,Dec,10,9:12:57,LabSZ,24507,reverse mapping checking getaddrinfo for customer-187-141-143-180-sta.uninet-ide.com.mx [187.141.143.180] failed - POSSIBLE BREAK-IN ATTEMPT!,E27,reverse mapping checking getaddrinfo for <*> [<*>] failed - POSSIBLE BREAK-IN ATTEMPT! host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log
>	11/14/25 9:12:59.000 AM	532,Dec,10,9:12:59,LabSZ,24507,Failed password for root from 187.141.143.180 port 35685 ssh2,E9,Failed password for <*> from <*> port <*> ssh2 host = TH3LILJ4NN47 source = CyberBlue Logs.csv sourcetype = cyberblue-log
>	11/14/25	534,Dec,10,9:13:03,LabSZ,24509,reverse mapping checking getaddrinfo for customer-187-141-143-180-sta.uninet-ide.com.mx [187.14

Attempts to log in via SSH were also made and the system could not recognize the usernames used for the actions.

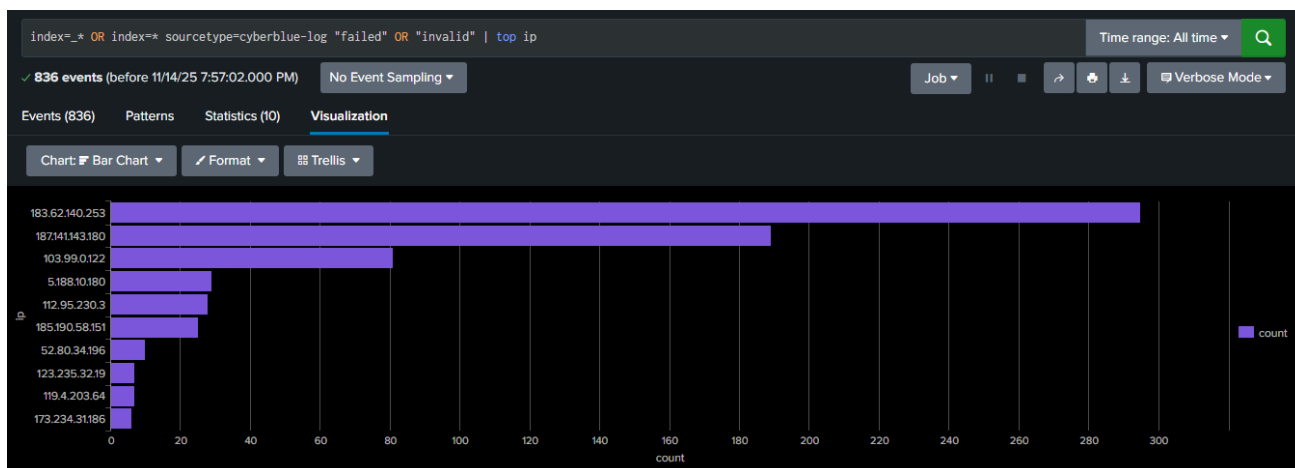


VISUALIZATIONS & ALERT

112.95.230.3 via port 11 was disconnected 20 times.



Visualized below are the IP addresses that attempted to authenticate and failed. They are categorized by count in descending order.



An alert was configured to run daily at 8am, expire after 24 hours, trigger when more than 2 results are generated and sent to changeemail@example@com in plain text format.

Daily Alert

GISOC

Enabled: Yes. [Disable](#)

App: search

Permissions: Private. Owned by th3ilj4nn47. [Edit](#)

Modified: Nov 14, 2025 8:32:21 PM

Alert Type: Scheduled. Daily, at 8:00. [Edit](#)

Trigger Condition: .. Number of Results is > 2. [Edit](#)

Actions: 1 Action [Edit](#)

[Send email](#)

RECOMMENDED ACTIONS

- Block high-frequency suspicious IPs (e.g., 183.62.140.253, 187.141.143.180) at the firewall or intrusion prevention system to prevent further automated authentication attempts.
- Enforce account lockout policies to stop repeated failed logins from being attempted on the same username.
- Enable and enforce on all externally accessible authentication endpoints.
- Restrict SSH access using measures like key-based authentication, disabling password authentication and allowing SSH only from a specific list.
- Increase log retention and forwarding to a SIEM, ensuring all authentication logs (including PAM, SSH, and application logs) are centrally monitored for brute-force activity.